

## Factoring out the greatest common factor (GCF)



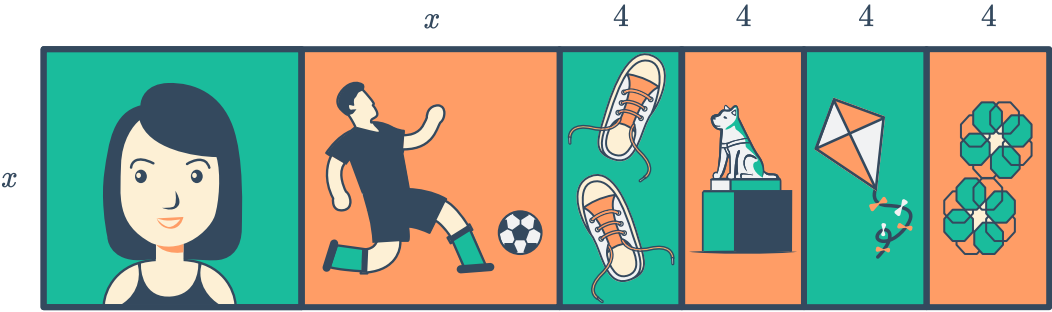
- 1
- a 3                                      b  $a$                                       c  $3x$                                       d  $3b$
- 2
- a  $4r + 12 = 4(r + 3)$                                       b  $5v - 30 = 5(v - 6)$   
c  $18c - 24 = 6(3c - 4)$                                       d  $2t^2 + 2t = 2t(t + 1)$
- 3
- a  $6(v + 5)$                                       b  $5(9t - 8)$                                       c  $-2(s + 5)$                                       d  $-2(6s - 5)$   
e  $y(y + 4)$                                       f  $k(k - 5)$                                       g  $2u(u - 4)$                                       h  $2t(2 + t)$   
i  $x(42 - x)$                                       j  $7(x + 5)$                                       k  $6(6 - x)$                                       l  $-2(x - 5)$   
m  $-10(10 + x)$                                       n  $-6(3y^2 - 1)$                                       o  $13(3x + 4)$                                       p  $(t - 3)(8t + 9)$

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### Additional questions

- 4 When factoring  $5xy$  out of the term  $-5xy$  you get  $\frac{-5xy}{5xy} = -1$  but Xander eliminated the term instead.
- 5
- a Let the width of the smallest field be  $x$  so that its area is  $x^2$ . The largest field has an area of  $9x^2$  and the middle field has an area of  $x(x + 5)$ .
- b The area of the entire field is the sum of each smaller field:  $x^2 + 9x^2 + x(x + 5)$ . The factored form of this expression is  $x(11x + 5)$  so the width of the field is  $x$  and the length is  $11x + 5$
- 6 When the greatest common factor is a monomial, we can use monomial division to determine what expression is left after the factoring is complete.
- 7 The greatest common factor represents the largest value that can be divided evenly from each term. If the greatest common factor of a variable had an exponent larger than any of the original terms, then dividing it out would leave a negative exponent. For example, consider the expression  $x^2 + x^3$ . If we divide both terms by  $x^2$  (the smaller exponent) then we are left with  $1 + x$ . If instead we try to divide both terms by  $x^3$  (the larger exponent) then we would be left with  $x^{-1} + 1$ .

a  $x(2x + 16)$



b  $2x(x + 8)$

